

Midterm Exam II

Statistics 1101

- ★ Write your name on each item.
 - ★ Answer the multiple choice questions on the provided sheet.
 - ★ Answer the short answer questions in your blue book.
 - ★ You may detach the stapled pages.
 - ★ Turn in this exam sheet with your blue book.
 - ★ Turn in your cheat sheet with your blue book.
 - ★ You may not use a calculator.
 - ★ You don't have to reduce fractions or simplify square roots.
- ✓ $\sqrt{\frac{10}{2}}$ and $\frac{1}{\sqrt{2}}$ are acceptable.
- ✗ $\frac{2+0}{1 \times 2}$ should be simplified to $\frac{2}{2}$ or 1.

Academic Integrity Agreement:

I understand what resources are and are not allowed on this exam. I won't cheat, discuss this exam with students who haven't yet taken the exam, or otherwise violate the university academic integrity standards. I will report any observed misconduct to the instructor or the appropriate authority as per university reporting procedures.

☐ I agree to the above. 🍷



Name:

Circle your answers on this page. Answer short answer questions in your blue book.

Select your exam version:    

1.) A B

2.) A B

3.) A B

4.) A B

5.) A B

6.) A B

7.) A B

8.) A B

9.) A B

10.) A B

11.) A B

12.) A B

13.) A B C D

14.) A B C D

15.) A B C D

16.) A B C D

17.) A B C D

18.) A B C D

19.) A B C D

20.) A B C D

21.) A B C D

22.) A B C D



CHEAT SHEET - STATS 1101 - SPRING 2024

Cumulative z -table (below)

z	+0.00	+0.01	+0.02	+0.03	+0.04	+0.05	+0.06	+0.07	+0.08	+0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633

Right-tailed t -test critical values

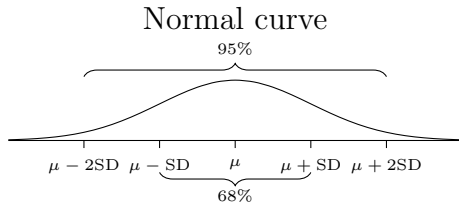
one-tail α d.f.	0.10	0.05	0.025	0.01	0.005
5	1.476	2.015	2.571	3.365	4.032
10	1.372	1.812	2.228	2.764	3.169
15	1.341	1.753	2.131	2.602	2.947
20	1.325	1.725	2.086	2.528	2.845
25	1.316	1.708	2.060	2.485	2.787
30	1.310	1.697	2.042	2.457	2.750
50	1.299	1.676	2.009	2.403	2.678
1000	1.282	1.646	1.962	2.330	2.581
∞	1.282	1.645	1.960	2.326	2.576

Pascal's triangle

Each row gives $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$.

$n = 0:$						1								
$n = 1:$						1		1						
$n = 2:$						1		2		1				
$n = 3:$						1		3		3		1		
$n = 4:$						1		4		6		4		1

Normal Curve



$$\text{Standardized Value} = \frac{x - \text{average}}{SD}$$

Probability

Conditional Probability:

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \text{ and } B)}{\mathbb{P}(B)}$$

General Multiplication Rule:

$$\mathbb{P}(A \text{ and } B) = \mathbb{P}(A | B)\mathbb{P}(B)$$

Addition Rule for Mutually Exclusive A & B:

$$\mathbb{P}(A \text{ or } B) = \mathbb{P}(A) + \mathbb{P}(B)$$

Independence: A and B are independent if

$$\mathbb{P}(A | B) = \mathbb{P}(A).$$

Bayes Theorem:

$$\begin{aligned} \mathbb{P}(A | B) &= \frac{\mathbb{P}(B | A)\mathbb{P}(A)}{\mathbb{P}(B)} \\ &= \frac{\mathbb{P}(B | A)\mathbb{P}(A)}{\mathbb{P}(B | A)\mathbb{P}(A) + \mathbb{P}(B | \text{not } A)\mathbb{P}(\text{not } A)}. \end{aligned}$$

Binomial Formula:

$$\mathbb{P}(k \text{ heads from } n \text{ tosses}) = \binom{n}{k} p^k (1-p)^{n-k}$$

Confidence Intervals

$$\bar{x} \pm z^* SE$$

where z^* is the critical value, depending on the confidence level.

Standard Errors

SE for sum of draws from a box:

$$\sqrt{n} \times \text{SD of box}$$

SE for average of draws from a box:

$$\frac{\text{SD of box}}{\sqrt{n}}$$

SE for sum of draws from a 0-1 box:

$$\sqrt{n} \times \sqrt{p(1-p)}$$

SE for sample proportion:

$$\sqrt{\frac{p(1-p)}{n}}$$

One-sided Hypothesis Testing

$$H_0 : \mu = \mu_0$$

$$H_A : \mu > \mu_0 \text{ (or } \mu < \mu_0)$$

Compare a **z-statistic** to a normal distribution.

$$z = \frac{\bar{x} - \mu}{SE}$$

Compare a **t-statistic** to a t distribution with $n-1$ degrees of freedom.

$$t = \frac{\bar{x} - \mu}{SE} \text{ where } SE = \frac{SD^+}{\sqrt{n}}$$

$$\begin{aligned} SD^+ &= \sqrt{\frac{1}{n-1} ((x_1 - \bar{x})^2 + \dots + (x_n - \bar{x})^2)} \\ &= \sqrt{\frac{n}{n-1}} \times SD \end{aligned}$$

The **P-value** is the probability (or area) of getting a result that is more extreme than the test statistic and in the direction allowed by the alternative hypothesis, assuming the null hypothesis is right. The **significance level** is the value α such that if the P-value is smaller, the null hypothesis is rejected. This corresponds to a confidence level of $1 - \alpha$.

You have exam version .

Following the instructions on the previous page counts for two points.

Multiple Choice (select one)

Two-choice questions are worth two points each. Four-choice questions are worth four points each. Select the best answer.

1. A bank wants to estimate the amount of change people carry. They take a simple random sample of 100 people and find that the average person carries \$0.50. They figure the standard error is \$0.05 because $\sqrt{\frac{0.5 \times 0.5}{100}} = 0.05$. Is this right?
 - A. Yes, it is correct.
 - B. No, it is wrong.
2. A researcher wants to estimate the height of students at a university. She obtains a simple random sample and calculates the average height of that sample. This value is an example of what?
 - A. parameter
 - B. statistic
3. A simple random sample is obtained by randomly selecting individuals from a population
 - A. with replacement
 - B. without replacement

Use the following information for the next two questions.

George randomly selects one of two coins with equal chance. The first coin is a trick coin that always comes up heads and the second always comes up tails.

4. After selecting a coin, George tosses the coin once. What is $\mathbb{P}(H)$?
 - A. 1
 - B. $\frac{1}{2}$
5. After selecting a coin, George tosses the coin twice. What is $\mathbb{P}(HH)$?
 - A. $\frac{1}{2}$
 - B. $\frac{1}{4}$
6. Fill in the blank: A P-value represents the probability of getting a result _____ than your calculated statistic, assuming that the null hypothesis is true.
 - A. more extreme
 - B. less extreme

7. Studies with the same P-value must have the same standard error.
 - A. True
 - B. False
8. Studies with the same P-value must have the same sample statistic.
 - A. True
 - B. False
9. The critical value for a right-tailed t -test with $n = 16$ and $\alpha = 0.05$ is
 - A. 1.645
 - B. 1.753
10. Felix is a junior statistician. He uses a z -test and fails to reject the null hypothesis. Felix's boss repeats the analysis with a t -test, using the same data and the same confidence level, but uses SD^+ instead of SD finds a critical value from a t distribution instead of from the normal distribution. What is true of the boss's analysis?
 - A. Also fail to reject
 - B. Ambiguous
11. We know the height of a certain population has an approximately normal distribution. We collect twelve observations and we want to use a hypothesis test to see if we should reject the hypothesis that the population average height is 5.5ft. Which hypothesis test should we use?
 - A. t -test
 - B. z -test
12. We collect n observations and we want to compare statistics used for the sum and the average of this data. The standard error of the average or proportion SE_{prop} is related to the standard error of the sum SE_{sum} by the following equation:
 - A. $SE_{prop} = \frac{1}{\sqrt{n}}SE_{sum}$
 - B. $SE_{prop} = \frac{1}{n}SE_{sum}$
13. A researcher sends out a survey, asking "How often do you take voluntary surveys." The researcher finds that 90% always take voluntary surveys. What kind of bias is at play?
 - A. non-response bias
 - B. response bias
 - C. chance error bias
 - D. non-chance error bias



Use the following information for the next two questions.

A company's shareholders elect its board of directors. Roberta the pollster surveys a simple random sample of 100 shareholders and find that 80 support candidate A.

14. If p describes what proportion of shareholders support candidate A, what is the SE related to the estimated $\hat{p}$?
- A. 4
 - B. .16
 - C. .064
 - D. .04
15. Roberta uses the above to find a confidence interval and declares that candidate A will win with very high confidence. However, she ignores that shareholders typically have multiple votes according to the number of shares they own. Her boss, Gloria, follows up with the 100 surveyed shareholders and finds that 99 of them have just one vote. The remaining shareholder does not support A and has 81 votes. Gloria constructs a new confidence interval for the proportion of *votes* that A will get. Gloria uses $n = 180$, corresponding to the total number of votes, finds $\hat{p} = \frac{4}{9} \approx 0.44$ and calculates the standard error as $\sqrt{\frac{\frac{4}{9} \times \frac{5}{9}}{180}} = \frac{1}{27}$. Is Gloria's analysis correct?
- A. Yes, because Gloria accounts for every vote.
 - B. No, because the 180 votes are not all independent.
 - C. No, because the central limit theorem will not hold if not everyone has an equal number of votes.
 - D. No, because Gloria miscalculated $\hat{p}$.
16. Which of the following is true?
- A. If the P-value is 0.05, the null hypothesis has only a 5% chance of being true.
 - B. A non-significant difference means the null is true.
 - C. If the null hypothesis is true and a 95% confidence level is used, a one-sided test will reject the null more often than a two-sided test.
 - D. If the null hypothesis is true and a 95% confidence level is used, we expect to reject the null hypothesis 5% of the time.
17. We flip a fair coin 4 times. The probability that the total number of heads is 2 is
- A. $\frac{1}{16}$
 - B. $\frac{2}{16}$
 - C. $\frac{4}{16}$
 - D. $\frac{6}{16}$



18. The *Columbia Versus Everyone Scholars Program* honors four students every year. Two students are selected from Columbia and the other two must be selected from the rest of the universities in the world. Suppose there are 4 qualifying students at Columbia and 6 in the rest of the world. How many possible selections can be made?
- A. More than $\binom{10}{4}$
 - B. $\binom{6}{2}$
 - C. $\binom{4}{2}$
 - D. Fewer than $\binom{10}{4}$

Use the following information for the next two questions.

We know a distribution has a standard deviation of 4 and we conduct a hypothesis test to see whether or not the mean is equal to 3. Let μ denote the mean, and we write $H_0 : \mu = 3$. We collect a sample of size 64 and observe the average is 6.

19. What is the test statistic?
- A. 6
 - B. $\frac{3}{24}$
 - C. 3
 - D. $-\frac{1}{6}$
20. What is the result of the test if $\alpha = 0.05$?
- A. Reject the null if H_A is $\mu > 3$
 - B. Reject the null if H_A is $\mu < 3$
 - C. Fail to reject the null if H_A is $\mu > 3$
 - D. Both A and B
21. In a simple random sample of 100 owls, the average lifespan is nine years and the SD is one year. What is a 50% confidence interval for the population average?
- A. 9 ± 0.1645
 - B. 9 ± 0.08225
 - C. 9 ± 0.0675
 - D. 9 ± 0.000
22. The critical value for a right-tailed t -test with $n = 19$ and $\alpha = 0.01$ is
- A. 0.504
 - B. 0.496
 - C. 2.528
 - D. Greater than 2.528 and less than 2.602
-



Short Answer

★ Show your work and justify your answers for full credit. ★

23. (4 points) The distribution of incomes in New York City is skewed, where the average is much higher than the median. One hundred different researchers conduct independent random samples with $n = 1000$ and find 100 sample averages. Sketch an appropriately shaped histogram for the different sample averages. Give a reason for the shape.
24. A gambler randomly selects one of two coins with equal chance. The coins are indistinguishable in appearance, but one is weighted so the probability of heads is 80% and the other is weighted so the probability of heads is 20%.
- (a) (2 points) What is the probability that the gambler gets a heads after one toss of the coin?
 - (b) (4 points) What is the probability that the gambler has the 80% coin if one toss is made and the result is a heads?
 - (c) (4 points) What is the probability of getting two heads from two tosses of the same randomly selected coin?
 - (d) (4 points) Now suppose that first, the gambler randomly selects one coin and gets a heads. Second, the gambler flips the other coin and gets a heads again. The gambler must make a third toss and they will win a jackpot if they get another heads. They can choose to use the first coin again or the second again. Which coin should they use? Or does it matter?
25. John buys a weighted coin. John is promised that the weighted coin will produce a heads more often. After $n = 36$ tosses of the coin, he finds a sample proportion $\hat{p} = \frac{24}{36}$ (or $\frac{4}{6}$) and conducts a one-sided hypothesis test with a significance level of 5%.
- (a) (2 points) State an appropriate alternative hypothesis if the null is $p = \frac{1}{2}$.
 - (b) (4 points) Find the appropriate standard error.
 - (c) (4 points) Find the test statistic z .
 - (d) (4 points) Find an approximate P-value.
 - (e) (2 points) Should John reject the null hypothesis?
26. A scientist is trying to study the movement patterns of bears in a single forest. The scientist tags bears that she finds at a specific river crossing and then studies them. Unbeknownst to the scientist, this is done at the beginning of winter, where bears that have already eaten enough have already begun to hibernate and are excluded from the sample.
- (a) (2 points) Describe a kind of bias in the study.
 - (b) (2 points) If the researcher wants to know the average distance covered per day by a bear during this time of the year, how might the sample average differ from the population average?

